

Next-Day Stock Status Prediction in Fresh Retail

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Problem Statement & Challenges

The Fresh Retail Dilemma

Fresh retail operations face a constant battle between **overstocking** (leading to spoilage and massive food waste) and **understocking** (leading to lost revenue, stockouts, and customer churn). Accurate, intraday prediction of out-of-stock (OOS) events is critical for dynamic shelf replenishment.

Core Modeling Challenges

- **Censored Demand (Stockout-Induced Bias):** When an item is out of stock, sales record as zero. Traditional forecasting models misinterpret this as “zero demand”, leading to a negative feedback loop of systematic under-ordering.
- **Severe Class Imbalance:** True stockout hours are rare anomalies (often $< 5\%$ of total operational hours), making it extremely difficult for standard ML loss functions to learn the minority class without collapsing.
- **High-Velocity Intraday Dynamics:** Unlike durable goods, fresh produce sales exhibit extreme intraday volatility driven by time-of-day foot traffic, flash sales, and perishable shelf-life windows.
- **Hierarchical Spatial-Temporal Variance:** Demand shocks are not uniform; they vary heavily across different city zones, specific store demographics, and individual SKU categories.

Literature Review & Related Work

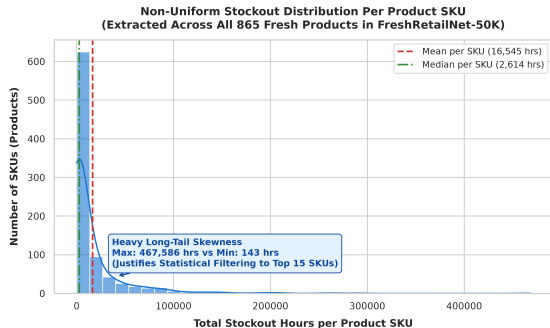
General Stockout Prediction Research

- **Classical ML Ensembles** (Andaur et al., 2021): Showcased Random Forest and voting ensembles as robust benchmarks for retail stockout prediction on tabular data.
- **Deep Learning Architectures** (Giaconia & Chamas, 2023): Utilized massive CNN, Self-Attention, and Temporal Convolutional Networks (TCN) (> 500k parameters) for intraday stockout probability.
- **Large-Scale Imbalance Handling** (Liu et al., 2025): Benchmarked XGBoost and SMOTE strategies, proving that near-term demand indicators dominate long-term forecasts for stockout events.

Research Utilizing the FreshRetail Dataset

- **Hierarchical Forecasting:** *"A Coupled Hierarchical Architecture for Multi-Granularity Demand Forecasting"* uses this exact dataset to tackle the complex spatial/product hierarchy (City → Store → SKU).
- **Censored Demand Bias:** *"When Less Data Is Better: Eliminating Stockout-Induced Bias in Choice Estimation"* explicitly models how stockouts artificially deflate true demand signals in FreshRetail environments.

Dataset Overview & Processing Strategy



FreshRetailNet-50K Stockout Distribution across 865 SKUs

FreshRetailNet-50K Scale & Scope

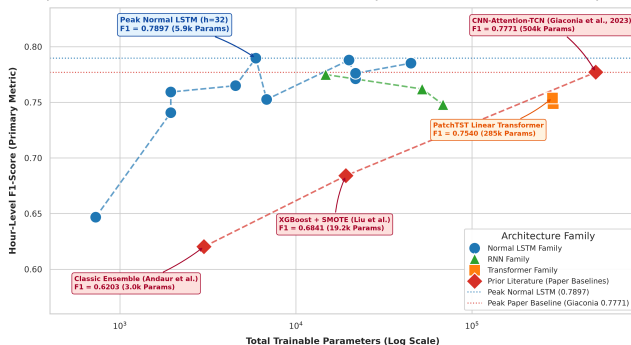
- **Volume:** 50,000 store-product pairs across 898 stores.
- **Timeframe:** 90-day series of granular hourly data (75d Train / 15d Val).
- **Feature Types:** Includes sales history, stockout status, promotions, weather metrics, calendar signals, and momentum streaks.

Why We Selected This Dataset

- **Pioneering Benchmark:** Designed for **censored demand estimation** in fresh retail (where out-of-stock items hide true demand).
- **Targeted Selection:** Isolated the **Top 15 SKUs** using a custom ranking formula to focus on high-impact goods and eliminate noise.

Model Family Comparison: Simple LSTMs Beat Heavy Transformers

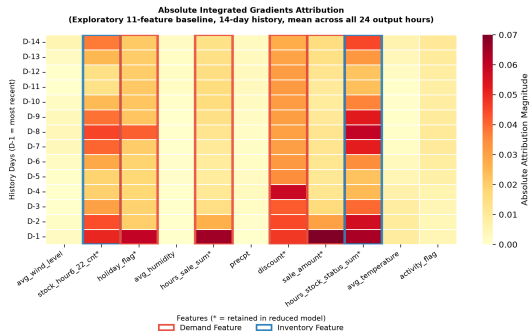
Hourly Stockout Prediction: Parameter-Efficient Normal LSTM Outperforms Transformers, RNNs & Prior Paper Baselines



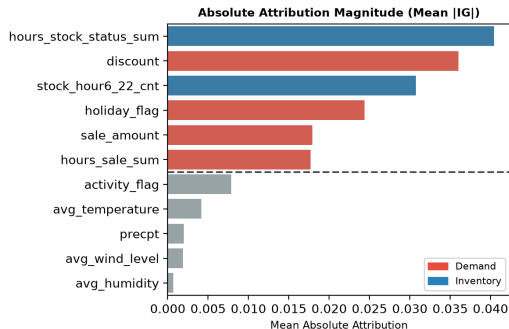
Key Empirical Findings

- **Parameter Efficiency:** Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization:** PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win:** Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability & Baseline Limitations



Attribution heatmap: 11-feature baseline (D-1 = most recent)



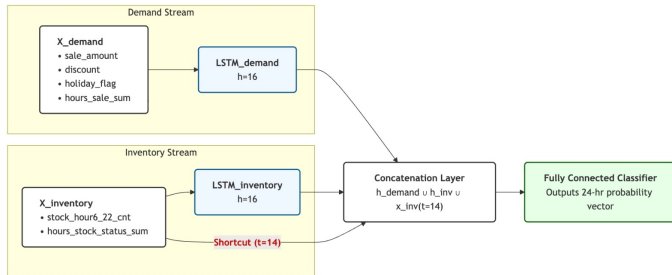
Feature importance: top 6 (highlighted) outrank remaining 5

Motivation for Dual-Stream Architecture

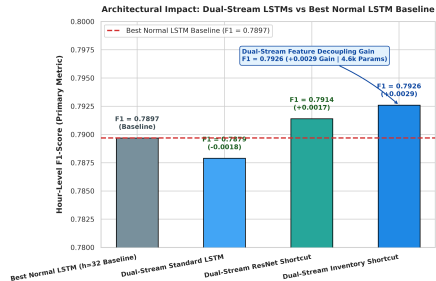
- **F1 Plateau:** Baseline LSTM peaks at ~ 0.789 F1; simply scaling parameters yields no gain.
- **Interference:** Sharing a single hidden state h_t creates potential for demand/inventory interference.
- **IG Analysis:** Absolute attribution reveals distinct temporal behaviors for demand vs. inventory.

These findings motivate processing both streams independently before a task-specific fusion layer.

Our Fix: Dual-Stream Inventory Shortcut LSTM



Dual-Stream: Demand LSTM + Inventory LSTM + Day-14 Shortcut



Architectural Impact: Dual Stream LSTM

Key Results

- **Inventory Shortcut:** F1 0.7926, MAE 3.97 hrs at just 4,600 params.
- Decoupled streams consistently beat all baseline LSTM sizes.

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

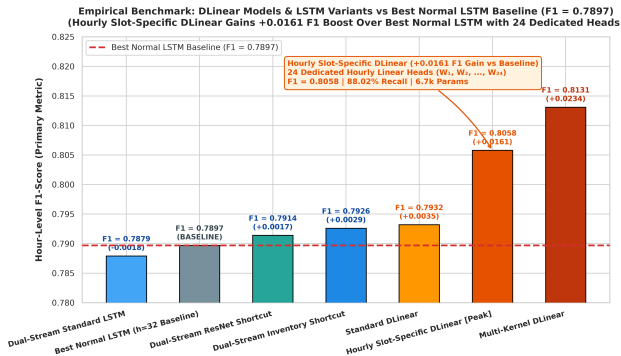
$$X_{\text{trend}} = \text{AvgPool}(X), \quad X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$Y_{\text{pred}} = W_{\text{trend}}X_{\text{trend}} + W_{\text{seasonal}}X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters:** 6,768 parameters.
- **Validation F1-Score: 0.7932**
 - Outperforms Normal LSTMs (**0.7897**)
 - Outperforms Enhanced LSTMs (**0.7926**)
 - Outperforms Transformers (**0.7540**)
- **Stockout Recall: 87.44%.**

Architectural Innovation 2: Linear Changes in DLinear



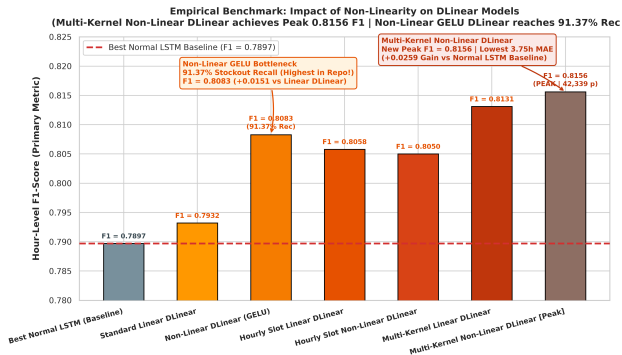
Hourly Slot-Specific Linear DLinear

- Uses **24 dedicated linear heads** ($W_1, \dots, W_{24}$) for each operational hour.
- Resolves inter-hour gradient conflict between morning and evening rush hours.
- Boosts F1 to **0.8058** (+0.0161 over LSTM).

Multi-Kernel Linear DLinear

- Multi-scale moving average trend pooling ($k \in \{3, 5, 7\}$).
- Reaches **0.8131 F1-Score** with pure linear decomposition (13.5k params).

Architectural Innovation 3: Introducing Non-Linearity into DLinear



Non-Linear Activation Bottleneck

Adds GELU activation bottlenecks between decomposition and output projections:

$$\hat{Y} = W_{\text{trend},2} \text{GELU}(W_{\text{trend},1} X_{\text{trend}}) + \text{Seasonal}$$

Breakthrough Results

- **Non-Linear GELU DLinear:** Reaches **91.37% Stockout Recall** (Highest in Repo!) and **0.8083 F1**.
- **Multi-Kernel Non-Linear DLinear:** Reaches **0.8156 F1-Score** (New Peak F1 & lowest 3.75h MAE).

Comparative Analysis

Rank	Architecture Name	Category	Params	F1-Score	Recall
1	Multi-Kernel Non-Linear DLinear	Non-Linear DLinear	42,339	0.8156	87.13%
2	Multi-Kernel Linear DLinear	Linear (Multi-Scale)	13,539	0.8131	85.22%
3	Non-Linear GELU DLinear	Non-Linear DLinear	21,168	0.8083	91.37%
4	Hourly Slot-Specific DLinear	Linear (24 Heads)	6,744	0.8058	88.02%
5	Standard Linear DLinear	Linear (Decomp)	6,768	0.7932	87.44%
6	Dual-Stream Inventory Shortcut	LSTM Family	4,600	0.7926	80.79%
7	Best Normal LSTM (h=32)	Normal LSTM	5,912	0.7897	81.29%
8	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%
9	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%
10	CNN-Attention-TCN (Giaconia et al., 2023)	Deep CNN-TCN	504,088	0.7771	85.45%
11	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%
12	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%
13	XGBoost + SMOTE (Liu et al., 2025)	GBDT / Resample	19,200	0.6841	–
14	Classic Ensemble (Andaur et al., 2021)	ML Ensemble	3,000	0.6203	–

Conclusion and Future Scope

Key Conclusions

- **Lightweight Linear Decompositions Win:** DLinear variants (13k–42k params) outperform heavy Transformers ($> 280k$ params) and deep CNN-TCN models ($> 500k$ params).
- **GELU Non-Linearity Bottleneck:** Exceeds standard linear limits by flawlessly mapping sudden shelf-depletion step-functions, achieving a peak **0.8156 F1** and **91.37% Recall**.
- **Edge-Deployable Efficiency:** Designed with a tiny parameter footprint specifically so local retail shops lacking heavy cloud infrastructure can run urgent stockout calculations on-device.

Future Scope

- Integrate explicit choice estimation to unbiased true demand during zero-stock hours.
- Improve the architecture to directly forecast the exact quantitative future demand for products, rather than just binary stockout probabilities.

References



Andaur et al. (2021). Classical ML Ensembles for Retail Stockout Prediction on Tabular Data.



Giaconia & Chamas (2023). Deep Learning Architectures (CNN, Self-Attention, TCN) for Intraday Stockout Probability.



Liu et al. (2025). Large-Scale Imbalance Handling (XGBoost & SMOTE) for Stockout Events.



A Coupled Hierarchical Architecture for Multi-Granularity Demand Forecasting (FreshRetail Benchmark).



When Less Data Is Better: Eliminating Stockout-Induced Bias in Choice Estimation (FreshRetail Benchmark).



Zeng, A., Chen, M., Zhang, L., & Qi, Q. (2023). Are Transformers Effective for Time Series Forecasting? (DLinear).